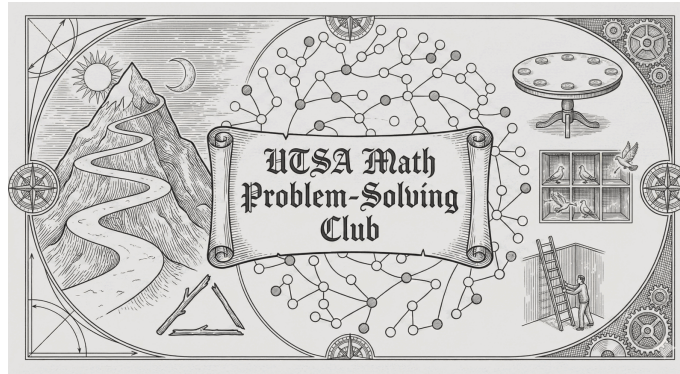




*“Mathematics is not a spectator sport.” — George Pólya*

### The Round Table

Two players take turns placing circular coins of equal size on a round table. The coins may touch the edge of the table or other coins, but they cannot overlap. The last player to be able to fit a coin on the table wins the game.

**Question:** If the table is large enough to hold at least one coin, which player has a winning strategy, and what is it?

### The Monk on the Mountain

A monk starts at the bottom of a mountain at sunrise and walks up a narrow path to the top, arriving at sunset. He stays at the top overnight. The next morning, he starts at the top at sunrise and walks down the exact same path, arriving at the bottom at sunset.

**Prove that:** There is at least one specific spot on the path that the monk occupied at the exact same time of day on both trips.

### The Broken Stick

A stick of unit length is broken at two points chosen independently and uniformly at random along the stick.

**Question:** What is the probability that the three resulting pieces can form a triangle?

### The 0-1 Multiple

**Prove that:** For any natural number  $n$ , there exists a non-zero multiple of  $n$  whose decimal representation contains only the digits 0 and 1.

*Hint: Consider the remainders of a specific sequence of numbers.*

### Arithmetic Mean of Digits

How many primes among the positive integers, written as usual in base 10, are such that their digits are alternating 1s and 0s, beginning and ending with 1?

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*Source: Putnam 1989 A1*

### Nilpotent Matrices

Let  $A$  be a square matrix such that  $A^k = 0$  for some integer  $k \geq 1$  (such a matrix is called *nilpotent*).

**Prove that:** The matrix  $(I - A)$  is invertible, and express its inverse as a polynomial in  $A$ .

### The “Impossible” Integral

Evaluate the following definite integral:

$$\int_0^{\pi/2} \frac{\sin^{100}(x)}{\sin^{100}(x) + \cos^{100}(x)} dx$$

*Hint: Do not even dream of finding antiderivatives!*

### The Hallway Corner

Two hallways, of widths  $a$  and  $b$ , meet at a right angle. What is the length of the longest ladder that can be carried horizontally around the corner?

### Integer Polynomials

Let  $P(x)$  be a polynomial with integer coefficients. Suppose that there exist three distinct integers  $a, b, c$  such that  $P(a) = P(b) = P(c) = -1$ .

**Prove that:**  $P(x)$  has no integer roots.

### Non-Consecutive Subsets

Let  $S = \{1, 2, 3, \dots, n\}$ . Determine the number of subsets of  $S$  that contain no two consecutive integers.

*Hint: Try small values of  $n$  and look for a pattern.*